

ABHILASH NEOG

🌐 [abhilash-neog.github.io](https://github.com/abhilash-neog) [in](#) LinkedIn [🎓](#) Scholar [G](#) Github [✉](mailto:abhilash22@vt.edu) abhilash22@vt.edu

Research Summary

Doctoral researcher working on generative AI and foundation models, with a focus on diffusion models, multimodal learning, and time-series modeling. My current work focuses on cross-modal alignment via flow matching, and agentic reasoning for knowledge-conditioned generation.

Education

Virginia Tech

Ph.D., Computer Science. Advisor: *Dr. Anuj Karpatne*. GPA: 4.0/4.0

Aug 2022 - May 2027

Blacksburg, USA

Virginia Tech

M.S., Computer Science. Advisor: *Dr. Anuj Karpatne*. GPA: 4.0/4.0

Aug 2022 - Dec 2024

Blacksburg, USA

Birla Institute of Technology and Science (BITS), Pilani

Bachelor of Engineering (B.E.), Computer Science, GPA: 8.08/10

July 2016 – July 2020

Pilani, India

Work Experience

Applied Scientist Intern | [Microsoft AI](#)

May 2026 – Aug 2026

- Proposed a training framework for flow-based generative models (FLUX) to preserve subject clarity in image outpainting, using LoRA fine-tuning with wavelet supervision, reducing FID by 7.7% over state-of-the-art approaches. *Paper under review*
- Developed an image outpainting pipeline to improve subject clarity, combining VLM-based semantic guidance, high-frequency patch estimation, and image injection for additional conditioning in FLUX models. *Patent submission under review*

Machine Learning Intern | [Kryptowire](#)

May 2023 – Aug 2023

- Developed an outlier detection model for denoising sensor-based Human Activity Recognition (HAR) **time series** data
- Built & deployed a CNN-based HAR model achieving 82% F-1 score on an android app using Keras & TensorFlow Lite

Data Scientist | [Oracle](#)

Sep 2020 – July 2022

- Built & deployed Machine Learning applications into ETL pipelines, leveraging Spark systems, MLOps & CI/CD pipelines
- Designed and deployed a *Demand Prediction* application for time series forecasting using the DeepAR model
- Developed an unsupervised classification algorithm achieving 40% higher accuracy than then LMs on a 71k-label dataset.

Software Engineer Intern | [VMware](#)

Jan 2020 – June 2020

- Streamlined Workspace ONE Cloud data workflows using Spring Boot REST APIs, and contributed to an end-user federation application with JUnit/Mockito testing

Software Engineer Intern | [Samsung Research Institute](#)

May 2019 – July 2019

- Optimized the C++ HAL call flow for Live Focus under low-light conditions, achieving considerable noise reduction during the first phase of camera capture

Research Experience

Graduate Research Assistant | [Knowledge-guided Machine Learning Lab](#)

Jan 2023 – Present

- Working on cross-modal alignment of LLMs & TS foundation models via flow matching for knowledge-conditioned generation
- Contributed to a novel physics-guided Diffusion Model for solving Partial Differential Eqns., achieving *15-250x* faster inference
- Investigated *zero-shot scene graph generation* using Vision-Language Models (VLMs) [*CVPRw 2025*]
- Developed a dual-scale attention framework for *fine-grained attribute grounding* in Large Vision-Language Models (LVLMs)
- Evaluated *reasoning capabilities* of VLMs on VQA tasks involving scientific images [*NeurIPS 2024*]
- Proposed a Model-agnostic Adaptation technique for Irregular and Sparse Time-Series (TS) modeling [*TMLR 2026*]
- Proposed an (a) *effective tokenization strategy* and (b) *Query-based forecasting*, to train a Foundation Model for spatio-temporal (probabilistic) predictions on irregular grids, leveraging multi-gpu training. [*KDD 2026*]

Talks/Awards

- Invited talk at the *ASLO 2026 Conference - Session on AI for the Next Frontier in Aquatic Sciences*
- Invited talk at the *AGU 2025 Conference - Session on Advancing Water Science Through AI*
- Talk on Transfer Learning in Lake Ecosystems at the “*NSF Macrosystems Biology Meeting 2024*”
- Lightning Talk at the “*Frontiers in Ecological Forecasting 2023*,” event at Virginia Tech
- NSF NAIRR (National AI Research Resource) Pilot Award, 2024*
- “*Star of the Month (Dec 2021)*” within the Oracle Analytics Cloud Organization, Oracle India

Publications

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen, Jian Jiao. “Preserving Subject-Clarity in Image Outpainting with Multiscale Wavelet Supervision”. *Under Review*
2. **Abhilash Neog**, Sepideh Fatemi, Medha Sawhney, et al. “LakeFM: Toward a Foundation Model for Aquatic Ecosystems Using Irregular Multivariate Multi-depth Time Series Data”. *KDD 2026* [pdf]
3. **Abhilash Neog**, Arka Daw, Sepideh Fatemi, Medha Sawhney, et al. “Investigating a Model-Agnostic and Imputation-Free Approach for Irregularly-Sampled Multivariate Time-Series Modeling”. *TMLR, 2026* [pdf]
4. A. Pandey, **Abhilash Neog**, Gautam Jajoo. “On the Internal Semantics of Time-Series Foundation Models”. *BERT2S @ NeurIPS 2025* [pdf]
5. M. Sawhney, **Abhilash Neog**, Mridul Khurana, Anuj Karpatne. “PRISMA: Improving the Accuracy–Latency Frontier of Diffusion-based PDE Solvers Using Physics-Informed Spectral Attention”. *Under review (ML4PS @ NeurIPS 2025)* [pdf]
6. S. Fatemi, **Abhilash Neog**, Amartya Dutta, M. Sawhney, et al. “Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance”. *CV4Science @ CVPR 2025 (Oral)*
7. A. Dutta, M. Sawhney, K.S. Mehrab, **Abhilash Neog**, Mridul Khurana, et al. “Open World Scene Graph Generation using Vision Language Models”. *World Models @ ICML 2025* [pdf]
8. KS Mehrab, M. Maruf, Arka Daw, **Abhilash Neog**, HB Manogaran, et al. “Fish-Vista: A Multi-Purpose Dataset for Understanding & Identification of Traits from Images”. *CVPR 2025* [pdf]
9. M. Maruf, Arka Daw, KS Mehrab, HB Manogaran, **Abhilash Neog**, M. Sawhney, et al. “VLM4Bio: A Benchmark Dataset to Evaluate Pretrained Vision-Language Models for Trait Discovery from Biological Images”. *NeurIPS 2024* [pdf]
10. Lavika Goel, **Abhilash Neog**, Ashish Aman, and Arshveer Kaur. “Hybrid Nature-Inspired Optimization Techniques in Face Recognition” *Transactions on Computational Science XXXVI, Springer LNCS, 2020*. [pdf]

Patents

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen. *Microsoft, 2026*. “Subject-Clarity-Aware Image Outpainting via Semantic and Structural Guidance with Vision-Language Models and High-Frequency Map Estimation.” *Patent submission (under review)*.
2. Akash B., Krishnan R., **Abhilash, N.**, Dipawesh P. and Karthik B. *Oracle, 2024*. “Machine Learning Based Spend Classification.” *U.S. Patent Application 17/903,161*. [pdf]

Academic Duties

- **Reviewing:**
 - Conferences/Journal - NeurIPS ('25, '26), AAAI ('26, '27), TMLR, ICML 2026, KDD 2026, ECCV 2026
 - Workshops - FM4Science (ICLR '26), FMSD (ICML '25, '26), FMs in the Wild (ICLR '25), Open Science for FMs (ICLR '25), XAI4Science (ICLR '25), ML4PS (NeurIPS '25), TSALM (NeurIPS '24)
- **Teaching:**
 - Graduate Teaching Assistant, CS 5805 Machine Learning, Spring 2024
 - Teaching Assistant, BITS F312 Neural Networks and Fuzzy Logic, Fall 2019

Technical Skills

Languages: Python, Java, C++, SQL, R

Frameworks: PyTorch, Git, Hydra, Tensorflow Keras, Spark